Consensus Control for Multi-Agent Systems under Switching Topologies via State-Dependent Dwell-Time Switching

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***Abstract*—This paper investigates the consensus problem for switched multi-agent systems with multiple candidate directed communication topologies. A consensus-orthogonal decomposition is first introduced to eliminate the redundant consensus component and transform the original multi-agent system into a reduced-order switched disagreement system. Based on this formulation, the strong connectivity of the union graph is used to exclude common non-consensus zero directions among the candidate topologies. A state-dependent dwell-time switching law is then designed via Lyapunov-Metzler inequalities to stabilize the disagreement error system. Under the jointly strongly connected condition, a strictly positive convex weight vector and a sufficiently large coupling parameter can be selected such that the associated zero-dwell-time lifted matrix is Hurwitz. Consequently, the Lyapunov-Metzler inequalities are feasible for every prescribed dwell time below an explicitly derived positive upper bound. It is shown that, if the Lyapunov-Metzler inequalities are feasible, the disagreement error converges to zero and global consensus is achieved. A numerical example is provided to illustrate the effectiveness of the proposed approach.**

***Index Terms*—Multi-agent consensus, switching topologies, jointly strongly connected, Lyapunov-Metzler inequalities, state-dependent dwell-time.**

I. Introduction

Consensus control is a fundamental problem in multi-agent systems, where a group of agents is required to reach agreement via local information exchange over communication networks [1], [2]. Owing to its broad applications in cooperative control, distributed sensing, formation control, unmanned systems, and networked autonomous systems, consensus has been extensively studied. In graph-based consensus analysis, connectivity of the communication topology is the standard condition for undirected networks [3]. For directed networks, typical graph-theoretic conditions require the communication graph to be strongly connected or to contain a directed spanning tree [4], [5]. These results reveal that sufficient information propagation across the network is essential for achieving consensus.

In practical networked systems, however, the communication topology may switch over time due to link failures, sensing limitations, communication constraints, or intermittent information exchange [3], [5]. Under such switching topologies, each instantaneous graph may be disconnected and may fail to guarantee global consensus by itself. Nevertheless, the collection of switching graphs may jointly provide sufficient information paths over time. This motivates the study of consensus under switching topologies with joint or union graph connectivity. Existing results have shown that consensus can still be achieved when the switching graphs satisfy jointly connected conditions, even though each instantaneous graph may not be connected [6]–[8].

Since switching topologies lead to a switched consensus dynamics, an effective way to analyze convergence is to separate the consensus component from the disagreement component. By using an orthogonal transformation, the original consensus problem can be converted into the stability problem of a reduced-order disagreement error system [9], [10]. The resulting reduced-order model contains only the disagreement component and can therefore be analyzed directly using switched-system stability tools.

The stability analysis of switched systems with possibly unstable subsystems provides another relevant perspective for studying switching-induced consensus dynamics. Classical switched-system theory has shown that a suitable switching law may stabilize the overall system even when some subsystem matrices are unstable or marginally stable [11]–[13]. More recent studies have developed mode-dependent dwell-time and multiple-Lyapunov-function methods for switched systems containing unstable modes [14]–[17]. These results motivate treating the reduced disagreement dynamics induced by individually disconnected topologies as a switched-system stabilization problem, in which consensus is achieved through an appropriately designed switching law rather than through the stability of every individual topology.

This paper addresses the consensus issue in switched multi-agent systems subject to multiple candidate directed communication topologies. Motivated by the state-dependent dwell-time switching framework proposed in [18], this paper applies a Lyapunov-Metzler-based dwell-time switching mechanism to the reduced-order disagreement dynamics of switched multi-agent systems with multiple candidate directed communication topologies. Each candidate topology is disconnected, whereas the union graph of all candidate topologies is assumed to be strongly connected. An orthogonal decomposition eliminates the redundant consensus component, and a state-dependent dwell-time switching law is designed to drive the disagreement state to zero, thereby achieving global consensus.

The main contributions are summarized as follows:

- A consensus-orthogonal decomposition is used to remove the redundant consensus direction and transform the original consensus problem into stabilization of a reduced-order switched disagreement system.
- A state-dependent switching law is developed based on finite-dwell-time Lyapunov-Metzler inequalities. It enforces a prescribed minimum dwell time and stabilizes the disagreement system without requiring the individual subsystem matrices to be Hurwitz.
- A constructive feasibility analysis is established using a lifted representation of the coupled Lyapunov operator. A Metzler matrix is constructed from the Hurwitz convex combination, and a Lyapunov-based block-matrix analysis together with a perturbation argument yields an explicit sufficient upper bound on the dwell time.

The remainder of the paper is organized as follows. Section II presents the graph preliminaries, system model, and reduced-order disagreement dynamics. Section III develops the state-dependent dwell-time switching law for the reduced linear switched system and proves its stability under feasible Lyapunov-Metzler inequalities. Section IV proves the existence of feasible Lyapunov-Metzler matrices from strong union connectivity and derives an explicit dwell-time bound. Section V presents numerical simulations. Section VI concludes the paper.

Notations: $\mathbb{R}^n$ denotes the n-dimensional Euclidean space, and $\mathbb{S}^{n\times n}$ denotes the space of real symmetric $n\times n$ matrices. The vector $\mathbf{1}_N \in \mathbb{R}^N$ stands for the N-dimensional all-ones vector. $I_n \in \mathbb{R}^{n\times n}$ is the n-order identity matrix, and $\mathbf{0}_n \in \mathbb{R}^n$ denotes the n-dimensional zero vector. For a symmetric matrix P, $P \succ 0$ and $P \prec 0$ denote positive and negative definiteness. The symbol $\otimes$ denotes the Kronecker product. For square matrices A and B, $A \oplus B = A\otimes I + I\otimes B$ denotes the Kronecker sum, with identity dimensions inferred from context. For vectors, $\|\cdot\|_2$ is the Euclidean norm; for matrices, it is the induced spectral norm. For a square matrix A, $\operatorname{spec}(A)$ denotes its spectrum. The spectral abscissa is defined as $\alpha(A) = \max_{\mu\in\operatorname{spec}(A)} \operatorname{Re}(\mu)$. The notation $\operatorname{col}\{\cdot\}$ denotes column stacking. For a matrix $X \in \mathbb{R}^{m\times n}$, $\operatorname{vec}(X) \in \mathbb{R}^{mn}$ denotes the column-wise vectorization obtained by stacking the columns of X. The class $\mathcal{M}$ contains Metzler matrices $\Pi = [\pi_{ij}] \in \mathbb{R}^{M\times M}$ satisfying $\pi_{ij} \geq 0$ for $i \neq j$ and $\Pi\mathbf{1}_M = \mathbf{0}_M$. The set of real symmetric matrices of dimension r is denoted by $\mathbb{S}^r$, and $\mathbb{S}^r_+ = \{Z \in \mathbb{S}^r : Z \succeq 0\}$ denotes the positive-semidefinite cone. The notation $(\mathbb{S}^r_+)^M$ denotes its M-fold Cartesian product. For a matrix Z, $\ker(Z)$ denotes its null space. For a closed convex set $\mathcal{K}$ and $Z \in \mathcal{K}$, $T_{\mathcal{K}}(Z)$ denotes the tangent cone to $\mathcal{K}$ at Z.

II. Preliminaries and Problem Formulation

Consider N agents with states $x_i(t) \in \mathbb{R}^d$. Define $x(t) = \operatorname{col}\{x_1(t), \ldots, x_N(t)\} \in \mathbb{R}^{Nd}$. The communication network switches among M candidate directed graphs $\mathcal{G}_p$. The adjacency weight $a^p_{ij} \geq 0$ characterizes the influence of agent j on agent i. Accordingly, $(j, i) \in \mathcal{E}_p$ if and only if $a^p_{ij} > 0$, while $a^p_{ij} = 0$ means no directed edge exists from node j to node i. The Laplacian matrix L_p is defined by $[L_p]_{ii} = \sum_{j\neq i} a^p_{ij}, \quad [L_p]_{ij} = -a^p_{ij},\ i \neq j$. It follows that $L_p\mathbf{1}_N = \mathbf{0}_N, p = 1, 2, \ldots, M$. A directed path from node i to node j consists of a sequence of directed edges connecting i to j. A directed graph is strongly connected if any node is reachable from every other node via a directed path.

Assumption 1: Each candidate directed topology $\mathcal{G}_p$ is disconnected, whereas the union graph $\mathcal{G}_U = \bigcup_{p=1}^M \mathcal{G}_p$ is strongly connected.

Under the active topology $\mathcal{G}_{\sigma(t)}$, where $\sigma(t) \in \Omega = \{1, \ldots, M\}$ is the switching signal, the agent dynamics under the standard consensus protocol are given by

$$\dot{x}_i(t) = -\sum_{j=1}^N a^{\sigma(t)}_{ij}\big(x_i(t) - x_j(t)\big). \tag{1}$$

By stacking $x(t) = \operatorname{col}\{x_1(t), \ldots, x_N(t)\}$, then (1) can be written in compact form as

$$\dot{x}(t) = -\big(L_{\sigma(t)} \otimes I_d\big)x(t), \tag{2}$$

where I_d is the d-dimensional identity matrix.

The multi-agent system achieves consensus if $\lim_{t\to\infty}\big(x_i(t) - x_j(t)\big) = \mathbf{0}_d$ for all $i, j \in \{1, 2, \ldots, N\}$, or equivalently when $x(t)$ converges to the consensus subspace $\mathcal{S}_c = \{\mathbf{1}_N \otimes \eta : \eta \in \mathbb{R}^d\}$.

To describe the disagreement among agents, define the centering matrix $H = I_N - \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top$. Choose $S \in \mathbb{R}^{(N-1)\times N}$ such that $S\mathbf{1}_N = \mathbf{0}_{N-1}$, $SS^\top = I_{N-1}$, $S^\top S = H$. Thus, the rows of S form an orthonormal basis of the subspace orthogonal to the consensus direction $\mathbf{1}_N$.

Define the reduced-order disagreement coordinate $\xi(t) = (S\otimes I_d)x(t) \in \mathbb{R}^r$, $r = (N-1)d$. Since $\operatorname{rank}(S) = N-1$ and $S1_N = \mathbf{0}_{N-1}$, we have $\ker(S) = \operatorname{span}\{\mathbf{1}_N\}$, which implies $\xi(t) = \mathbf{0}_r$ if and only if there exists $\eta(t) \in \mathbb{R}^d$ such that $x(t) = \mathbf{1}_N\otimes\eta(t)$. Hence, $\xi(t) \to \mathbf{0}_r$ is equivalent to consensus of the original multi-agent system.

Define the arithmetic-average coordinate $\eta(t) = \frac{1}{N}(\mathbf{1}_N^\top \otimes I_d)x(t) \in \mathbb{R}^d$. Using $I_N = \frac{1}{N}\mathbf{1}_N\mathbf{1}_N^\top + S^\top S$, the original state can be decomposed as

$$x(t) = \mathbf{1}_N \otimes \eta(t) + (S^\top \otimes I_d)\xi(t). \tag{3}$$

Differentiating $\xi(t)$ and using (2) and (3) gives

$$\begin{aligned}\dot{\xi}(t) &= -(SL_{\sigma(t)} \otimes I_d)x(t)\\ &= -\big(SL_{\sigma(t)}\mathbf{1}_N \otimes I_d\big)\eta(t) - \big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t)\\ &= -\big(SL_{\sigma(t)}S^\top \otimes I_d\big)\xi(t).\end{aligned} \tag{4}$$

Define $\bar{A}_p = -\left(SL_pS^\top \otimes I_d\right)$. Then, the reduced-order disagreement dynamics are represented by the switched linear system

$$\dot{\xi}(t) = \bar{A}_{\sigma(t)}\xi(t). \tag{5}$$

III. State-Dependent Dwell-Time Switching Design for the Reduced Linear System

Consider the reduced-order disagreement system (5).

Define the disagreement output as $z(t) = C\xi(t) \in \mathbb{R}^q$, where $C \in \mathbb{R}^{q\times r}$, $q \geq r$, is a given full-column-rank matrix. In particular, $C = I_r$ corresponds to $q = r$ and gives direct measurement of the complete disagreement state.

For each mode $i \in \Omega$, choose a symmetric positive-definite matrix $X_i = X_i^\top \in \mathbb{R}^{r\times r} \succ 0$ and define $V_i(\xi) = \xi^\top X_i\xi$. Let $\Pi = [\pi_{ij}] \in \mathcal{M}$ be a Metzler matrix satisfying $\pi_{ij} \geq 0$, for $i \neq j$, and $\sum_{j=1}^{M} \pi_{ij} = 0$. For a prescribed minimum dwell time $T > 0$, define

$$Y_{1,j} = e^{\bar{A}_j^\top T} X_j e^{\bar{A}_j T},\ Y_{2,j} = \int_0^T e^{\bar{A}_j^\top \tau} C^\top C e^{\bar{A}_j \tau}\, d\tau. \tag{6}$$

The finite-dwell-time Lyapunov-Metzler inequalities are given by

$$\bar{A}_i^\top X_i + X_i\bar{A}_i + \sum_{j\neq i} \pi_{ij}\left(Y_{1,j} + Y_{2,j} - X_i\right) + C^\top C \prec 0. \tag{7}$$

For a given active mode $i = \sigma(t_k)$, define $\Omega_i = \Omega \setminus \{i\}$. For every $j \in \Omega_i$, let

$$\begin{aligned} \Delta_{ij}(t) &= \xi^\top(t)\left(Y_{1,j} + Y_{2,j} - X_i\right)\xi(t), \\ J_j(t) &= \xi^\top(t)\left(Y_{1,j} + Y_{2,j}\right)\xi(t). \end{aligned} \tag{8}$$

The next switching instant is defined as

$$t_{k+1} = \inf\left\{t > t_k + T : \exists j \in \Omega_i,\ \Delta_{ij}(t) < 0\right\}, \tag{9}$$

and the active topology is selected according to

$$\begin{aligned} \sigma(t) &= i, && t \in [t_k, t_{k+1}), \\ \sigma(t_{k+1}) &= \arg\min_{j\in\Omega_i} J_j(t_{k+1}). \end{aligned} \tag{10}$$

By construction, the switching law satisfies $t_{k+1} - t_k \geq T$, and therefore excludes Zeno behavior.

Based on the above switching law, the following result is a direct specialization of the state-dependent dwell-time stabilization framework in [18] to the reduced-order disagreement system (5).

Theorem 1: Suppose that there exist matrices $X_i = X_i^\top \succ 0$, $i \in \Omega$, and a Metzler matrix $\Pi \in \mathcal{M}$ satisfying the Lyapunov–Metzler inequalities (7). Then the switching law (9)–(10) renders the disagreement system (5) asymptotically stable. In particular, $\lim_{t\to\infty} \xi(t) = \mathbf{0}_r$. Consequently, the original multi-agent system achieves global consensus, that is,

$$\lim_{t\to\infty}\left(x_i(t) - x_j(t)\right) = \mathbf{0}_d, \qquad \forall i, j \in \{1, \dots, N\}.$$

The validity of the above theorem follows directly from [18, Theorem 2], and is therefore omitted here.

Remark 1: The inequalities in (7) do not require every individual subsystem matrix $\bar{A}_i$ to be Hurwitz. A disconnected candidate topology may leave some disagreement directions nondecaying when it acts alone. Consensus is achieved through the coordinated state-dependent switching among the candidate topologies rather than through the asymptotic stability of each individual mode.

IV. Existence and Construction of a Metzler Matrix

This section shows that, under Assumption 1, the condition $Q = C^\top C \succ 0$, and a sufficiently small prescribed dwell time, one can construct a Metzler matrix such that the finite-dwell-time Lyapunov-Metzler inequalities are feasible.

Choose any strictly positive convex weight vector $\lambda = [\lambda_1, \dots, \lambda_M]^\top \in \mathbb{R}^M$, $\lambda_p > 0$, $\sum_{p=1}^{M} \lambda_p = 1$. Define $L_\lambda = \sum_{p=1}^{M} \lambda_p L_p$, we obtain

$$\bar{A}_\lambda = \sum_{p=1}^{M} \lambda_p \bar{A}_p = -\left(SL_\lambda S^\top \otimes I_d\right).$$

Lemma 1: Under Assumption 1, $\bar{A}_\lambda$ is Hurwitz for every strictly positive convex weight vector λ.

Proof: Since $\lambda_p > 0$ and all adjacency weights are nonnegative, the graph associated with L_λ has the same edge set as the union graph. Hence, it is strongly connected.

Define $\mathcal{T} = \begin{bmatrix}\mathbf{1}_N & S^\top\end{bmatrix}$, $\mathcal{T}^{-1} = \begin{bmatrix}\frac{1}{N}\mathbf{1}_N^\top \\ S\end{bmatrix}$. Using $L_\lambda\mathbf{1}_N = \mathbf{0}_N$, one obtains $\mathcal{T}^{-1}L_\lambda\mathcal{T} = \begin{bmatrix}0 & \frac{1}{N}\mathbf{1}_N^\top L_\lambda S^\top \\ \mathbf{0}_{(N-1)\times 1} & SL_\lambda S^\top\end{bmatrix}$. Therefore,

$$\operatorname{spec}(L_\lambda) = \operatorname{spec}(\mathcal{T}^{-1}L_\lambda\mathcal{T}) = \{0\} \cup \operatorname{spec}(SL_\lambda S^\top).$$

Since the graph associated with L_λ is strongly connected, the zero eigenvalue of L_λ is simple and all its remaining eigenvalues have strictly positive real parts. Hence, all eigenvalues of $SL_\lambda S^\top$ have strictly positive real parts. Consequently,

$$\bar{A}_\lambda = -\left(SL_\lambda S^\top \otimes I_d\right)$$

is Hurwitz. ■

For each mode $i \in \Omega$, define $E_i(T) = e^{\bar{A}_iT}$ and $\mathcal{B}_i = \bar{A}_i^\top \oplus \bar{A}_i^\top = \bar{A}_i^\top \otimes I_r + I_r \otimes \bar{A}_i^\top \in \mathbb{R}^{r^2\times r^2}$. Furthermore, let $\mathscr{B} = \operatorname{diag}\{\mathcal{B}_1, \dots, \mathcal{B}_M\}$, and $\Pi_d = \operatorname{diag}\{\pi_{11}, \dots, \pi_{MM}\}$. For a matrix tuple $\mathbf{X} = (X_1, \dots, X_M)$, define the homogeneous linear operator $\mathfrak{F}_T$ componentwise by

$$[\mathfrak{F}_T(\mathbf{X})]_i = \bar{A}_i^\top X_i + X_i\bar{A}_i + \pi_{ii}X_i + \sum_{j\neq i} \pi_{ij}E_j^\top(T)X_jE_j(T). \tag{11}$$

Using $\operatorname{vec}(AXB) = (B^\top \otimes A)\operatorname{vec}(X)$ and $E_j^\top(T) \otimes E_j^\top(T) = e^{\bar{A}_j^\top T} \otimes e^{\bar{A}_j^\top T} = e^{\mathcal{B}_jT}$, one obtains

$$\begin{aligned} &\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right) \\ &= \left[I_r \otimes \bar{A}_i^\top + \bar{A}_i^\top \otimes I_r + \pi_{ii}I_{r^2}\right]\operatorname{vec}(X_i) \\ &\quad + \sum_{j\neq i} \pi_{ij}\left(E_j^\top(T) \otimes E_j^\top(T)\right)\operatorname{vec}(X_j) \\ &= (\mathcal{B}_i + \pi_{ii}I_{r^2})\operatorname{vec}(X_i) + \sum_{j\neq i} \pi_{ij}e^{\mathcal{B}_jT}\operatorname{vec}(X_j). \end{aligned} \tag{12}$$

Stacking (12) for $i = 1, \dots, M$ yields

$$\operatorname{col}\left\{\operatorname{vec}\left([\mathfrak{F}_T(\mathbf{X})]_i\right)\right\}_{i=1}^{M} = \mathcal{J}_T \operatorname{col}\left\{\operatorname{vec}(X_i)\right\}_{i=1}^{M}, \tag{13}$$

where the (i,j)th block of $\mathcal{J}_T$ is

$$[\mathcal{J}_T]_{ij} = \begin{cases} \mathcal{B}_i + \pi_{ii} I_{r^2}, & i = j, \\ \pi_{ij} e^{\mathcal{B}_j T}, & i \neq j. \end{cases}$$

Since $e^{\mathscr{B}T} = \operatorname{diag}\left\{e^{\mathcal{B}_1 T}, \dots, e^{\mathcal{B}_M T}\right\}$, the above block matrix can be written compactly as

$$\mathcal{J}_T = \mathscr{B} + \Pi_d \otimes I_{r^2} + [(\Pi - \Pi_d) \otimes I_{r^2}]\, e^{\mathscr{B}T}. \tag{14}$$

Lemma 2: Given a Metzler matrix $\Pi \in \mathcal{M}$, a prescribed dwell time $T > 0$, and a matrix C satisfying $Q = C^\top C \succ 0$, define $\mathcal{J}_T$ by (14). If $\mathcal{J}_T$ is Hurwitz, then the Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1, \dots, M.$$

Proof: Define $W_i(T) = Q + \sum_{j \neq i} \pi_{ij} Y_{2,j}(T)$. Since $Y_{2,j}(T) = \int_0^T e^{\bar{A}_j^\top \tau} Q e^{\bar{A}_j \tau} d\tau \succ 0$, it follows that $W_i(T) \succ 0$. Using $\pi_{ii} = -\sum_{j \neq i} \pi_{ij}$, the Lyapunov-Metzler inequalities can be written as

$$[\mathfrak{F}_T(\mathbf{X})]_i + W_i(T) \prec 0, \qquad i = 1, \dots, M. \tag{15}$$

Suppose that $\mathcal{J}_T$ is Hurwitz. Consider the auxiliary matrix-valued linear system

$$\dot{\mathbf{Z}}(s) = \mathfrak{F}_T\big(\mathbf{Z}(s)\big), \qquad \mathbf{Z}(s) = \big(Z_1(s), \dots, Z_M(s)\big), \tag{16}$$

Since $\mathfrak{F}_T$ maps symmetric matrix tuples into symmetric matrix tuples, $(\mathbb{S}^r)^M$ is invariant under (16).

Define the stacked vector $\boldsymbol{\zeta}(s) = \operatorname{col}\left\{\operatorname{vec}\big(Z_i(s)\big)\right\}_{i=1}^M \in \mathbb{R}^{Mr^2}$. By (13), system (16) is equivalent to

$$\dot{\boldsymbol{\zeta}}(s) = \mathcal{J}_T \boldsymbol{\zeta}(s).$$

Hence, $\boldsymbol{\zeta}(s) = e^{\mathcal{J}_T s}\boldsymbol{\zeta}(0)$. Since $\mathcal{J}_T$ is Hurwitz, this finite-dimensional linear system is exponentially stable. Therefore, there exist constants $c > 0$ and $\gamma > 0$ such that

$$\|\boldsymbol{\zeta}(s)\|_2 \le c e^{-\gamma s} \|\boldsymbol{\zeta}(0)\|_2, \qquad s \ge 0. \tag{17}$$

In particular, $\boldsymbol{\zeta}(s) \to \mathbf{0}_{Mr^2}$ as $s \to \infty$, and therefore $Z_i(s) \to \mathbf{0}_{r\times r}$ for all $i = 1, \dots, M$.

Define $\widetilde{A}_i = \bar{A}_i + \frac{\pi_{ii}}{2} I_r$. Then the ith component of (16) is

$$\dot{Z}_i = \widetilde{A}_i^\top Z_i + Z_i \widetilde{A}_i + \sum_{j \neq i} \pi_{ij} E_j^\top(T) Z_j E_j(T). \tag{18}$$

We now verify the tangent-cone condition for $(\mathbb{S}_+^r)^M$. Let $\mathbf{Z} = (Z_1, \dots, Z_M) \in (\mathbb{S}_+^r)^M$ and take any $v \in \ker(Z_i) \subset \mathbb{R}^r$. Since $Z_i v = \mathbf{0}_r$,

$$v^\top \dot{Z}_i v = \sum_{j \neq i} \pi_{ij} \big(E_j(T) v\big)^\top Z_j \big(E_j(T) v\big) \ge 0, \tag{19}$$

where the inequality follows from $\pi_{ij} \ge 0$ and $Z_j \succeq 0$.

According to [19], the tangent cone $\mathbb{S}_+^r$ at Z_i is given by

$$T_{\mathbb{S}_+^r}(Z_i) = \left\{H \in \mathbb{S}^r : v^\top H v \ge 0,\ \forall v \in \ker(Z_i)\right\}.$$

Hence, by (19) and the tangent-cone rule for Cartesian products,

$$\mathfrak{F}_T(\mathbf{Z}) \in T_{(\mathbb{S}_+^r)^M}(\mathbf{Z}), \qquad \forall \mathbf{Z} \in (\mathbb{S}_+^r)^M.$$

Since $(\mathbb{S}_+^r)^M$ is a closed convex cone, the Nagumo invariance theorem [20] implies

$$\mathbf{Z}(0) \in (\mathbb{S}_+^r)^M \implies \mathbf{Z}(s) \in (\mathbb{S}_+^r)^M, \qquad s \ge 0.$$

Choose an arbitrary scalar $\rho > 1$ and define $R_i = \rho W_i(T) \succ 0$. Let $\mathbf{R} = (R_1, \dots, R_M)$ and let $\mathbf{Z}(s; \mathbf{R}) = \big(Z_1(s; \mathbf{R}), \dots, Z_M(s; \mathbf{R})\big)$ denote the solution of (16) with $\mathbf{Z}(0; \mathbf{R}) = \mathbf{R}$. Define $\mathbf{r} = \operatorname{col}\left\{\operatorname{vec}(R_i)\right\}_{i=1}^M \in \mathbb{R}^{Mr^2}$. The associated stacked solution satisfies $\operatorname{col}\left\{\operatorname{vec}\big(Z_i(s; \mathbf{R})\big)\right\}_{i=1}^M = e^{\mathcal{J}_T s}\mathbf{r}$. Therefore, by (17),

$$\left\|\operatorname{col}\left\{\operatorname{vec}\big(Z_i(s; \mathbf{R})\big)\right\}_{i=1}^M\right\|_2 \le c e^{-\gamma s}\|\mathbf{r}\|_2.$$

Hence, every component $Z_i(s; \mathbf{R})$ is integrable on $[0, \infty)$.

For each $i \in \Omega$, define

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds, \tag{20}$$

and let $\mathbf{X} = (X_1, \dots, X_M)$. Since $\mathbf{R} \in (\mathbb{S}_+^r)^M$ and the cone $(\mathbb{S}_+^r)^M$ is forward invariant, $Z_i(s; \mathbf{R}) \succeq 0, s \ge 0$. Therefore,

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds \succeq 0, \qquad i = 1, \dots, M.$$

In particular, $X_i = X_i^\top$.

To establish strict positive definiteness, observe that $Z_i(0; \mathbf{R}) = R_i = \rho W_i(T) \succ 0$. Since $Z_i(s; \mathbf{R})$ is continuous with respect to s, there exists $\delta_i > 0$ such that $Z_i(s; \mathbf{R}) \succ 0, 0 \le s \le \delta_i$. Consequently,

$$X_i = \int_0^\infty Z_i(s; \mathbf{R})\, ds \ \succeq \int_0^{\delta_i} Z_i(s; \mathbf{R})\, ds \ \succ 0.$$

Thus,

$$X_i = X_i^\top \succ 0.$$

Finally, since $\mathfrak{F}_T$ is a finite-dimensional linear operator and the above integral is convergent, $\mathfrak{F}_T$ can be moved inside the integral. Using $\dot{\mathbf{Z}}(s; \mathbf{R}) = \mathfrak{F}_T\big(\mathbf{Z}(s; \mathbf{R})\big)$, one obtains

$$\begin{aligned} \mathfrak{F}_T(\mathbf{X}) &= \mathfrak{F}_T\left(\int_0^\infty \mathbf{Z}(s; \mathbf{R})\, ds\right) \\ &= \int_0^\infty \mathfrak{F}_T\big(\mathbf{Z}(s; \mathbf{R})\big) ds \\ &= \int_0^\infty \dot{\mathbf{Z}}(s; \mathbf{R}) ds \\ &= \lim_{s \to \infty} \mathbf{Z}(s; \mathbf{R}) - \mathbf{Z}(0; \mathbf{R}) \\ &= -\mathbf{R}. \end{aligned}$$

Therefore, for every $i = 1, \dots, M$,

$$\begin{aligned} [\mathfrak{F}_T(\mathbf{X})]_i + W_i(T) &= -R_i + W_i(T) \\ &= -\rho W_i(T) + W_i(T) \\ &\prec 0. \end{aligned}$$

Hence, the Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions. ■

Define $\Pi_0 = \mathbf{1}_M \lambda^\top - I_M$, $\Pi_\kappa = \kappa \Pi_0$, $\kappa > 0$. Since $\lambda > 0$ entrywise, Π_κ is Metzler. Using $\lambda^\top \mathbf{1}_M = 1$, it follows that

$$\begin{aligned} \Pi_\kappa \mathbf{1}_M &= \kappa\big(\mathbf{1}_M \lambda^\top \mathbf{1}_M - \mathbf{1}_M\big) = \mathbf{0}_M, \\ \lambda^\top \Pi_\kappa &= \kappa\big(\lambda^\top \mathbf{1}_M \lambda^\top - \lambda^\top\big) = \mathbf{0}_M^\top. \end{aligned}$$

For $T = 0$, $e^{\mathscr{B}T} = I_{Mr^2}$, whence (14) simplifies to

$$\mathcal{J}_0(\Pi_\kappa) = \mathscr{B} + \Pi_\kappa \otimes I_{r^2}. \tag{21}$$

Lemma 3: Under Assumption 1, there exists $\kappa^\star > 0$ such that $\mathcal{J}_0(\Pi_\kappa)$ is Hurwitz for every $\kappa > \kappa^\star$.

Proof: Π_0 has a simple zero eigenvalue with right eigenvector $\mathbf{1}_M$ and left eigenvector λ, satisfying $\Pi_0\mathbf{1}_M = \mathbf{0}_M$ and $\lambda^\top\Pi_0 = \mathbf{0}_M^\top$. Furthermore, for every $y \in \ker(\lambda^\top) \subset \mathbb{R}^M$, $\Pi_0 y = \mathbf{1}_M(\lambda^\top y) - y = -y$. Since $\dim\ker(\lambda^\top) = M-1$, one obtains $\sigma(\Pi_0) = \left\{0, \underbrace{-1,\dots,-1}_{M-1 \text{ times}}\right\}$.

Choose $\Phi \in \mathbb{R}^{M\times(M-1)}$ whose columns form a basis of $\ker(\lambda^\top)$, and define $U_M = \begin{bmatrix}\mathbf{1}_M & \Phi\end{bmatrix}$. Since $\lambda^\top\mathbf{1}_M = 1$ and $\lambda^\top\Phi = \mathbf{0}_{M-1}^\top$, the matrix U_M is nonsingular and the first row of U_M^{-1} is $\lambda^\top$. It follows that

$$\begin{aligned}\Pi_0 U_M &= \begin{bmatrix}\Pi_0\mathbf{1}_M & \Pi_0\Phi\end{bmatrix} = \begin{bmatrix}\mathbf{0}_M & -\Phi\end{bmatrix}\\ &= U_M\begin{bmatrix}0 & \mathbf{0}_{M-1}^\top\\ \mathbf{0}_{(M-1)\times 1} & -I_{M-1}\end{bmatrix}.\end{aligned}$$

which yields $U_M^{-1}\Pi_0 U_M = \mathrm{diag}\{0, -I_{M-1}\}$.

Define $U = U_M \otimes I_{r^2}$. Partition $U^{-1}\mathscr{B}U = \begin{bmatrix}\bar{\mathcal{B}} & \mathcal{B}_{12}\\ \mathcal{B}_{21} & \mathcal{B}_{22}\end{bmatrix}$, where $\bar{\mathcal{B}} = (\lambda^\top \otimes I_{r^2})\mathscr{B}(\mathbf{1}_M \otimes I_{r^2}) = \sum_{i=1}^M \lambda_i\mathcal{B}_i$. Moreover, $U^{-1}(\Pi_\kappa \otimes I_{r^2})U = \mathrm{diag}\left\{0, -\kappa I_{(M-1)r^2}\right\}$. Therefore,

$$U^{-1}\mathcal{J}_0(\Pi_\kappa)U = \begin{bmatrix}\bar{\mathcal{B}} & \mathcal{B}_{12}\\ \mathcal{B}_{21} & \mathcal{B}_{22} - \kappa I_{(M-1)r^2}\end{bmatrix}. \tag{22}$$

Using $\mathcal{B}_i = \bar{A}_i^\top \oplus \bar{A}_i^\top$, one obtains

$$\bar{\mathcal{B}} = \sum_{i=1}^M \lambda_i\left(\bar{A}_i^\top \oplus \bar{A}_i^\top\right) = \bar{A}_\lambda^\top \oplus \bar{A}_\lambda^\top.$$

By Lemma 1, $\bar{A}_\lambda$ is Hurwitz. Therefore, $\bar{\mathcal{B}}$ is Hurwitz.

Let $P_\mathcal{B} = P_\mathcal{B}^\top \succ 0$ be the unique solution of $\bar{\mathcal{B}}^\top P_\mathcal{B} + P_\mathcal{B}\bar{\mathcal{B}} = -I_{r^2}$. Define $\mathcal{P} = \mathrm{diag}\left\{P_\mathcal{B}, I_{(M-1)r^2}\right\} \succ 0$ and $\mathcal{H} = P_\mathcal{B}\mathcal{B}_{12} + \mathcal{B}_{21}^\top$. Denote the transformed matrix in (22) by $\widehat{\mathcal{J}}_0(\kappa)$. Then

$$\widehat{\mathcal{J}}_0^\top(\kappa)\mathcal{P} + \mathcal{P}\widehat{\mathcal{J}}_0(\kappa) = \begin{bmatrix}-I_{r^2} & \mathcal{H}\\ \mathcal{H}^\top & \mathcal{B}_{22}^\top + \mathcal{B}_{22} - 2\kappa I_{(M-1)r^2}\end{bmatrix}.$$

Choose $\kappa^\star > 0$ sufficiently large such that

$$2\kappa^\star > \lambda_{\max}\left(\mathcal{B}_{22}^\top + \mathcal{B}_{22} + \mathcal{H}^\top\mathcal{H}\right).$$

Then, for every $\kappa > \kappa^\star$,

$$\mathcal{B}_{22}^\top + \mathcal{B}_{22} - 2\kappa I_{(M-1)r^2} + \mathcal{H}^\top\mathcal{H} \prec 0.$$

Since the upper-left block is $-I_{r^2} \prec 0$, the Schur-complement condition gives

$$\widehat{\mathcal{J}}_0^\top(\kappa)\mathcal{P} + \mathcal{P}\widehat{\mathcal{J}}_0(\kappa) \prec 0.$$

Hence, $\widehat{\mathcal{J}}_0(\kappa)$ is Hurwitz. Because $\widehat{\mathcal{J}}_0(\kappa)$ is similar to $\mathcal{J}_0(\Pi_\kappa)$, it follows that $\mathcal{J}_0(\Pi_\kappa)$ is Hurwitz for every $\kappa > \kappa^\star$. ■

Theorem 2: Suppose that Assumption 1 holds and $Q = C^\top C \succ 0$. Choose a strictly positive convex vector $\lambda = [\lambda_1,\dots,\lambda_M]^\top$, $\sum_{i=1}^M \lambda_i = 1$, $\lambda_i > 0$. Let $\kappa > \kappa^\star$, where $\kappa^\star$ is given by Lemma 3, and define $\Pi = \kappa\left(\mathbf{1}_M\lambda^\top - I_M\right)$. Let $\mathcal{J}_0 = \mathcal{J}_0(\Pi) = \mathscr{B} + \Pi \otimes I_{r^2}$, and let $P = P^\top \succ 0$ be the unique solution of $\mathcal{J}_0^\top P + P\mathcal{J}_0 = -I_{Mr^2}$. Define

$$T_{\mathrm{ub}} = \frac{1}{\|\mathscr{B}\|_2}\ln\left(1 + \frac{1}{2\|P\|_2\|\Pi - \Pi_d\|_2}\right). \tag{23}$$

Then, for every prescribed dwell time satisfying $0 < T < T_{\mathrm{ub}}$, the finite-dwell-time Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions

$$X_i = X_i^\top \succ 0, \qquad i = 1,\dots,M.$$

Proof: Let $\Delta_T = \mathcal{J}_T - \mathcal{J}_0$. By (14) and the definition of $\mathcal{J}_0$, $\Delta_T = [(\Pi - \Pi_d)\otimes I_{r^2}]\left(e^{\mathscr{B}T} - I_{Mr^2}\right)$.

By the submultiplicativity of the spectral norm and the identity $\|A\otimes B\|_2 = \|A\|_2\|B\|_2$,

$$\begin{aligned}\|\Delta_T\|_2 &\le \|(\Pi - \Pi_d)\otimes I_{r^2}\|_2\|e^{\mathscr{B}T} - I_{Mr^2}\|_2\\ &= \|\Pi - \Pi_d\|_2\|e^{\mathscr{B}T} - I_{Mr^2}\|_2.\end{aligned} \tag{24}$$

Using $e^{\mathscr{B}T} - I_{Mr^2} = \int_0^T \mathscr{B}e^{\mathscr{B}s}ds$, we derive

$$\|e^{\mathscr{B}T} - I_{Mr^2}\|_2 \le \int_0^T \|\mathscr{B}\|_2 e^{\|\mathscr{B}\|_2 s}ds = e^{\|\mathscr{B}\|_2 T} - 1. \tag{25}$$

Combining (24) and (25) yields

$$\|\Delta_T\|_2 \le \|\Pi - \Pi_d\|_2\left(e^{\|\mathscr{B}\|_2 T} - 1\right). \tag{26}$$

Since $\mathcal{J}_T = \mathcal{J}_0 + \Delta_T$,

$$\begin{aligned}\mathcal{J}_T^\top P + P\mathcal{J}_T &= \mathcal{J}_0^\top P + P\mathcal{J}_0 + \Delta_T^\top P + P\Delta_T\\ &= -I_{Mr^2} + \Delta_T^\top P + P\Delta_T.\end{aligned} \tag{27}$$

For any $y \in \mathbb{R}^{Mr^2}$, $y^\top(\Delta_T^\top P + P\Delta_T)y = 2y^\top P\Delta_T y \le 2\|P\|_2\|\Delta_T\|_2\|y\|_2^2$, thus

$$\Delta_T^\top P + P\Delta_T \preceq 2\|P\|_2\|\Delta_T\|_2 I_{Mr^2}.$$

Substituting into (27) yields

$$\mathcal{J}_T^\top P + P\mathcal{J}_T \preceq -\left[1 - 2\|P\|_2\|\Delta_T\|_2\right]I_{Mr^2}. \tag{28}$$

A sufficient condition for negative definiteness is $2\|P\|_2\|\Delta_T\|_2 < 1$. By (26), this is guaranteed by

$$2\|P\|_2\|\Pi - \Pi_d\|_2\left(e^{\|\mathscr{B}\|_2 T} - 1\right) < 1,$$

which rearranges to

$$T < \frac{1}{\|\mathscr{B}\|_2}\ln\left(1 + \frac{1}{2\|P\|_2\|\Pi - \Pi_d\|_2}\right) \triangleq T_{\mathrm{ub}}.$$

Thus, for all $0 < T < T_{\mathrm{ub}}$, $\mathcal{J}_T^\top P + P\mathcal{J}_T \prec 0$. By the Lyapunov stability theorem, $\mathcal{J}_T$ is Hurwitz. By Lemma 2, the finite-dwell-time Lyapunov-Metzler inequalities (7) admit symmetric positive-definite solutions. ■

V. Numerical Simulations

Consider a multi-agent system with $N = 4$ agents, $d = 1$, and two candidate topologies shown in Fig. 1. Each topology is disconnected, whereas their union graph is strongly connected.

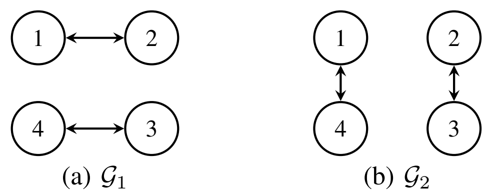

Fig. 1. Candidate communication topologies.

The corresponding reduced matrices satisfy $\sigma(\bar{A}_1) = \sigma(\bar{A}_2) = \{-2, -2, 0\}$, and hence neither subsystem is Hurwitz. For $\lambda = [0.5, 0.5]^\top$, the convex combination satisfies $\sigma(\bar{A}_\lambda) = \{-2, -1, -1\}$, which verifies Lemma 1.

Choose $\Pi = \begin{bmatrix} -0.27 & 0.27 \\ 0.27 & -0.27 \end{bmatrix}$, $C = I_3$. Theorem 2 gives $T_{\mathrm{ub}} = 0.1647$ s. Accordingly, the prescribed dwell time is selected as $T = 0.1$ s $< T_{\mathrm{ub}}$, for which the Lyapunov-Metzler inequalities are feasible.

For $x(0) = [5, -3, 4, 2]^\top$, $\sigma(0) = 1$, the simulation results are shown in Figs. 2 and 3.

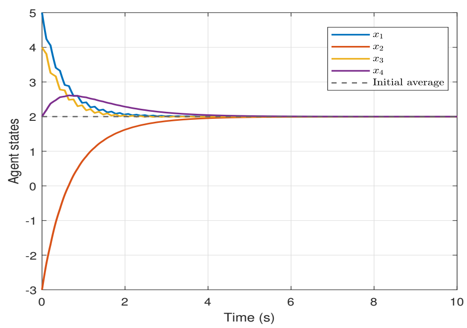

Fig. 2. Agent state trajectories under the proposed switching law.

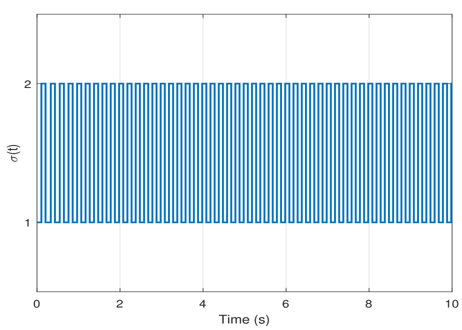

Fig. 3. State-dependent switching signal.

The agent states converge to the common value 2, while every switching interval is no less than the prescribed dwell time $T = 0.1$ s. Therefore, consensus is achieved although neither candidate topology can independently guarantee global consensus.

VI. Conclusion

This paper studied the consensus problem for multi-agent systems under switching directed topologies. A consensus-orthogonal decomposition was used to transform the original problem into stabilization of a reduced-order switched disagreement system. A state-dependent switching law with a prescribed minimum dwell time was developed based on Lyapunov-Metzler inequalities. Under strong connectivity of the union graph, a Hurwitz convex combination of the reduced subsystem matrices was obtained and used to construct a Metzler matrix. A lifted representation and a perturbation argument were then employed to establish feasibility of the Lyapunov-Metzler inequalities and derive an explicit sufficient upper bound on the dwell time. Numerical results demonstrated that consensus can be achieved even when none of the individual subsystem matrices is Hurwitz.

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